$$L\frac{\partial P}{\partial L}+K\frac{\partial P}{\partial K} = 3L\left(\frac{K}{L}\right)^{\frac{1}{4}}+K\left(\frac{L}{K}\right)^{\frac{3}{4}}$$

$$= 3L^{\frac{3}{4}}K^{\frac{1}{4}}+L^{\frac{3}{4}}K^{\frac{1}{4}}$$

$$= 4L^{\frac{3}{4}}K^{\frac{1}{4}} = P$$

Hence Euler's theorem is verified.

Example 6.43

The demand for a commodity x is $q = 5-2p_1+p_2-p_1^2p_2$. Find the partial elasticities $\frac{Eq}{EP_1}$ and $\frac{Eq}{EP_2}$ when p_1= 3 and p_2=7.

Solution:

$$\frac{\partial q}{\partial p_1} = -2-2p_1p_2$$

$$\frac{\partial q}{\partial p_2} = 1-p_1^2$$

(i) $$\frac{Eq}{Ep_1} = -\frac{P_1}{q}\frac{\partial q}{\partial p_1}$$

$$= \frac{-p_1}{5-2p_1+p_2-p_1^2p_2}(-2-2p_1p_2)$$

$$= \frac{2p_1+2p_1^2p_2}{5-2p_1+p_2-p_1^2p_2}$$

When p_1= 3 and p_2= 7

$$\frac{Eq}{Ep_1} = \frac{2(3)+2(9)(7)}{5-6+7-(9)(7)} = \frac{132}{-57} = \frac{-132}{57}$$

(ii) $$\frac{Eq}{Ep_2} = -\frac{P_2}{q}\frac{\partial q}{\partial p_2} = \frac{-p_2(1-p_1^2)}{5-2p_1+p_2-p_1^2p_2}$$

$$= \frac{-p_2+p_2p_1^2}{5-2p_1+p_2-p_1^2p_2}$$

When p_1= 3 and p_2= 7

$$\frac{Eq}{Ep_2} = \frac{-7+7(9)}{5-6+7-(9)(7)} = \frac{56}{-57} = \frac{-56}{57}$$

Exercise 6.5

1. Find the marginal productivities of capital (K) and labour (L) if $P = 8L-2K+3K^2-2L^2+7KL$ when K = 3 and L= 1.
2. If the production of a firm is given by $P = 4LK-L^2+K^2$, $L > 0$, $K > 0$, Prove that $L\frac{\partial P}{\partial L}+K\frac{\partial P}{\partial K} = 2P$.
3. If the production function is $z = 3x^2-4xy+3y^2$ where x is the labour and y is the capital, find the marginal productivities of x and y when x=1, y=2.
4. For the production function $P = 3(L)^{0.4}(K)^{0.6}$, find the marginal productivities of labour (L) and capital (K) when L = 10 and K = 6.
[use: $(0.6)^{0.6}$ =0.736, $(1.67)^{0.4}$=1.2267].
5. The demand for a quantity A is $q = 13-2p_1-3p_2^2$. Find the partial elasticities $\frac{Eq}{Ep_1}$ and $\frac{Eq}{Ep_2}$ when p_1= p_2=2.
6. The demand for a commodity A is $q = 80-p_1^2+5p_2-p_1p_2$. Find the partial elasticities $\frac{Eq}{Ep_1}$ and $\frac{Eq}{Ep_2}$ when p_1=2, p_2=1.

Exercise 6.6

Choose the Correct answer:

1. Average fixed cost of the cost function $C(x)$= $2x^3+5x^2-14x+21$ is
(a) $\frac{2}{3}$ (b) $\frac{5}{x}$ (c) $-\frac{14}{x}$ (d) $\frac{21}{x}$
2. Marginal revenue of the demand function p= 20–3x is
(a) 20–6x (b) 20–3x
(c) 20+6x (d) 20+3x

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